

NAFNet-Alpha: Controllable Smartphone Image Denoising with Adjustable Grain Retention

Feature-wise Linear Modulation for user-controlled denoising · Real Smartphone Photos · SIDD Dataset

42.60

dB PSNR ($\alpha=0$)

7.96

dB Spread

20.38M

Model size

INTRODUCTION

Low-light smartphone photos often contain strong sensor noise. Automatic noise reduction can remove noise, but may over-smooth fine details. Most denoising models produce only one fixed output for each noisy image. This project builds a controllable NAFNet-based denoising model. A single alpha (α) parameter adjusts the balance between clean denoising and texture retention. The control is learned inside the network, not added as a simple post-processing blend. In short, the model restores noisy smartphone images while letting the user choose a cleaner or more textured result.

ALPHA CONTROL MODES

Mode	α Value	Effect
Clean	0.00	Maximum denoising
Natural	0.20–0.30	Balanced texture
Grain	0.50–1.00	Retained film-like texture

Output formula: $\hat{y} = x - r \times (1.0 - 0.8\alpha) \rightarrow$ At $\alpha=1$ still denoises 20%.

MOTIVATION & EXISTING CHALLENGES

- Real sensor noise:** SIDD captures multi-ISO smartphone noise — far harder than synthetic Gaussian.
- Fixed denoising:** NAFNet (39.97 dB), Restormer (40.02 dB) — competitive PSNR but zero grain control.
- BM3D over-smoothing:** Destroys texture at high ISO (12500), no fine-grained control possible.
- DnCNN domain gap:** SSIM $\approx$ 1 on test = near-identity output. Not usable on real smartphone noise.
- No single-model control:** Prior work blends post-hoc. We solve this in one network via FiLM conditioning.

OUR 6 CONTRIBUTIONS

- NAFBlocks**
Depthwise separable conv + SimpleGate + SCA. GELU-free, no InstanceNorm.
- AlphaFiLM (5 $\times$)**
FiLM at bottleneck + all 4 decoder levels. Per-channel scale & shift on α .
- AlphaGatedSkip**
Skip connections gated by α . Controls encoder texture flow to decoder.
- Output Scaling**
 $\hat{y}=x-r \cdot (1-0.8\alpha)$ — guaranteed mathematical α effect, always active.
- Capped Target**
 $t(\alpha)=(1-0.45\alpha)\cdot\text{clean} + 0.45\alpha\cdot\text{noisy}$. Teaches natural grain, not artifacts.
- Teacher Preservation**
Frozen $\alpha=0$ checkpoint as teacher. Prevents degradation during fine-tuning.

DESIGN CHANGES FROM EARLIER MODELS

Component	Old Approach	NAFNet-Alpha
Block type	Conv U-Net	NAFBlock U-Net
Skip connections	Direct skip	Alpha-gated skip
Alpha injection	Post-process blend	FiLM at 5 locations
Identity risk ($\alpha=1$)	Returns raw noisy input	Denoising floor (0.8 $\times$)
Supervision	Clean image only	Capped + teacher loss

DATA DESCRIPTION

Training: SIDD — 30,000 patches 256 $\times$ 256 from 320 real noisy/clean pairs. SIDD only, no mixing.
Validation: 1,000 held-out patches. Metrics: PSNR + SSIM.
Real-photo testing: personal smartphone at ISO 3,500–12,500.
GPU: NVIDIA L4 (Google Colab).

POSTER TAKEAWAYS

- Clean ($\alpha=0$):**
42.60 dB on our 1k SIDD validation subset; adds alpha control.
- Natural ($\alpha=0.2-0.3$):**
Balanced denoising with visible texture. Recommended default.
- Grain ($\alpha=1$):**
34.64 dB — natural film-like texture. 7.96 dB lower than clean.

BASELINE MODEL COMPARISON

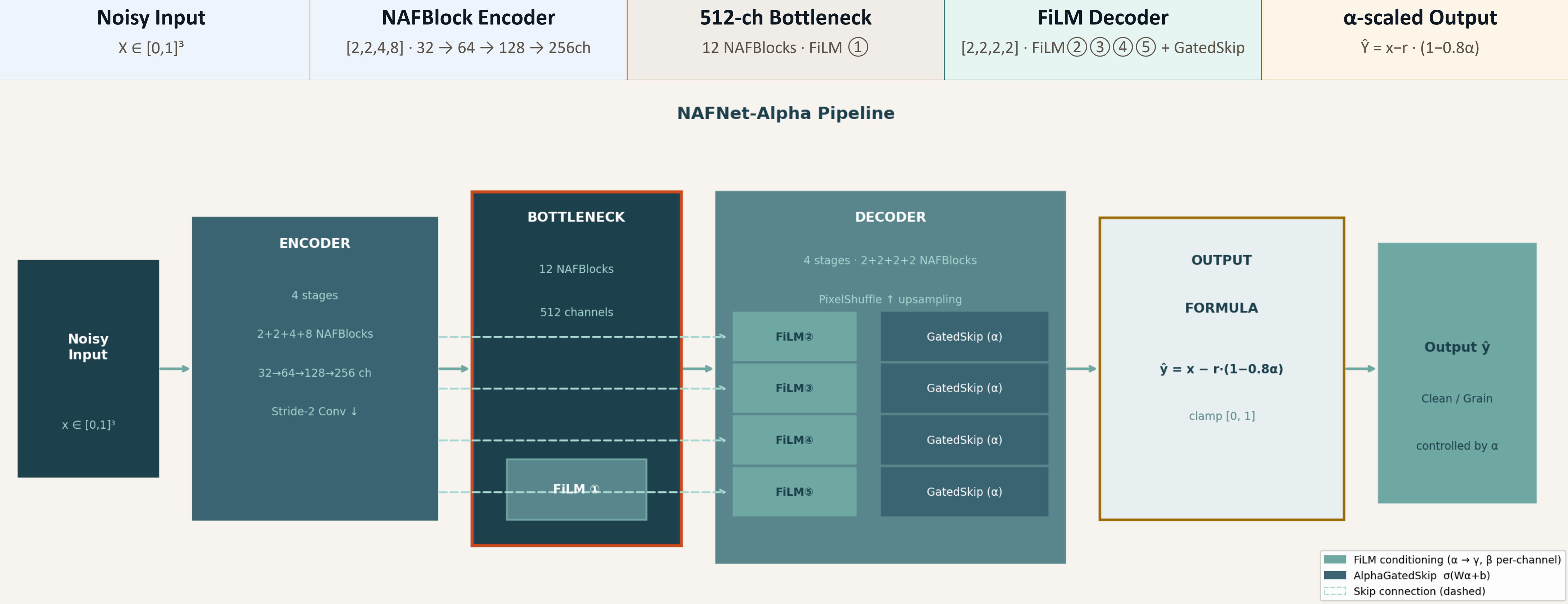


ISO 4000 · Center crop · PSNR: NAFNet $\alpha=0$: 31.48 dB · NAFNet $\alpha=1$: 36.74 dB · BM3D: 31.19 dB · DnCNN: SSIM=0.99 (domain gap)



ISO 12500 · Full scene · PSNR: NAFNet $\alpha=0$: 19.37 dB · NAFNet $\alpha=1$: 24.76 dB · BM3D: 24.71 dB · DnCNN: SSIM=0.99 (domain gap)

METHODOLOGY — ARCHITECTURE



FiLM conditioning at 5 locations (bottleneck + 4 decoder levels). AlphaGatedSkip on all skip connections. Output formula guarantees denoising floor at all α .

ARCHITECTURE DETAIL CARDS

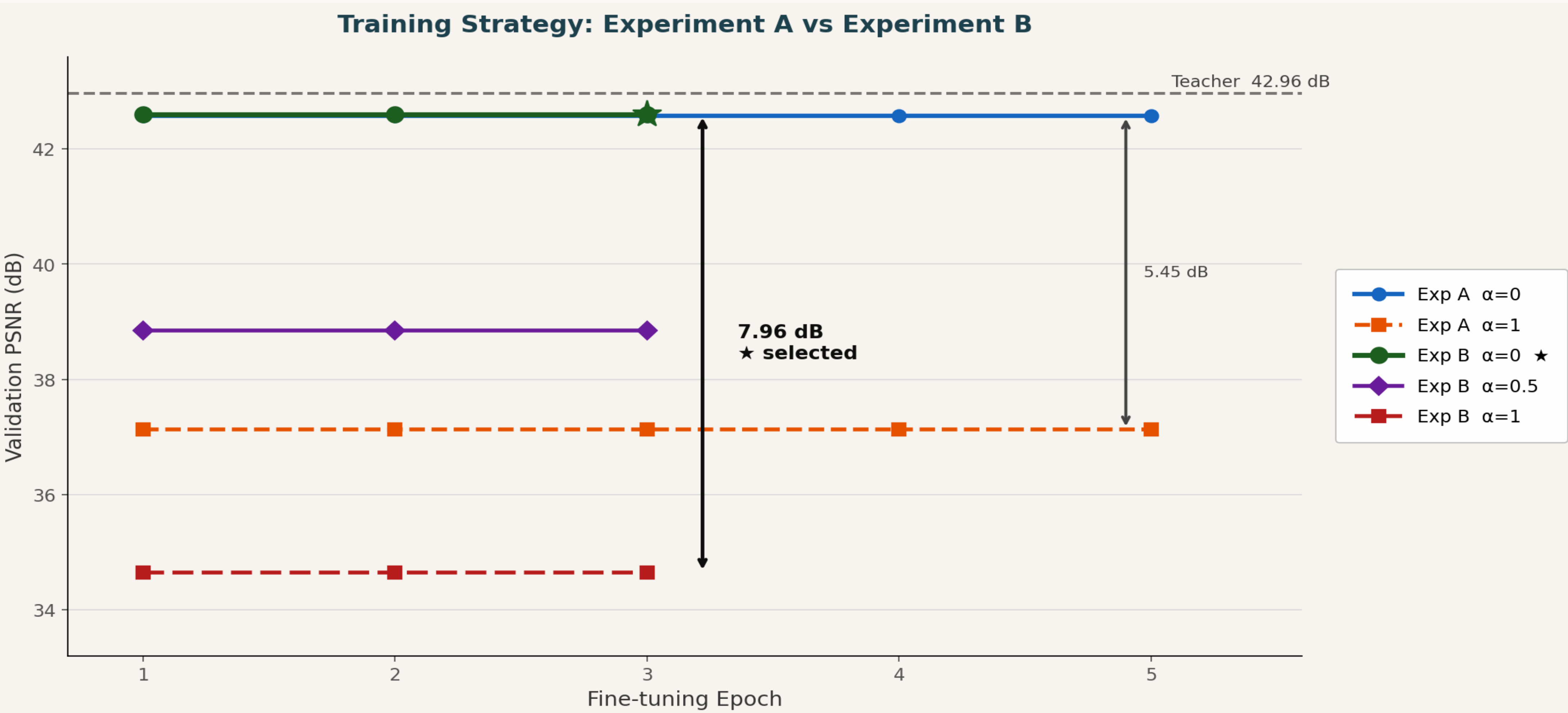
NAFBlock	AlphaFiLM
LayerNorm $\rightarrow$ depthwise 3 $\times$ 3 $\rightarrow$ SimpleGate $\rightarrow$ SCA $\rightarrow$ pointwise. Residual + β, γ init to 0 — identity at training start.	MLP(α) $\rightarrow \gamma, \beta$ per channel $\cdot x \cdot (1+\tanh(\gamma))+\beta$. Applied at bottleneck + 4 decoder levels (5 total).
AlphaGatedSkip	Model Specs
skip $\times \alpha(W\alpha+b)$ — controls encoder texture flow to decoder via learned differentiable gate.	Width: 32 · Peak: 512ch · 20,379,939 params · 5 FiLM modules · GELU-free · no InstanceNorm

LOSS & FINE-TUNING DETAILS

$L = L_1 + 0.01 \cdot L_{FFT} + 1.0 \cdot L_{teacher}$		
L_1	L_{FFT}	$L_{teacher}$
Pixel reconstruction. Main denoising objective.	Frequency-domain L1. Preserves fine texture detail.	Frozen $\alpha=0$ checkpoint. Prevents quality loss.
Capped target: $t(\alpha) = (1-0.45\alpha) \cdot y_{clean} + 0.45\alpha \cdot x_{noisy} \rightarrow \alpha=0$: clean $\cdot \alpha=1$: 55% clean + 45% noisy (natural grain)		

TRAINING STRATEGY

Stage 1 · Base Train	Exp A · CAP=0.30	★ Exp B · FINAL
42.96 dB PSNR · SIDD val Trained from scratch · No alpha	5.45 dB spread · LR=2e-5 · 5 epochs $\alpha=0$: 42.58 · $\alpha=1$: 37.13 · subtle	7.96 dB spread · LR=1e-5 · cap=0.45 $\alpha=0$: 42.60 · $\alpha=1$: 34.64 ✓



Exp B (cap=0.45) achieves 7.96 dB spread vs 5.45 dB in Exp A. Teacher model (42.96 dB) preserved throughout. Higher ALPHA_CAP forces meaningful grain retention.

EXPERIMENTAL RESULTS — PSNR

Clean	Balanced	Grain
42.60 $\alpha = 0.0$ Max denoising	38.85 $\alpha = 0.5$ Natural texture	34.64 $\alpha = 1.0$ Texture retained

7.96 dB Spread $\alpha=0 \rightarrow \alpha=1$. Visible in every tested scene. Alpha is intrinsic to network features — not a post-processing blend.

SIDD PSNR COMPARISON VS PUBLISHED METHODS

Method	Type	PSNR	Ctrl?
BM3D	Classical	25.65 dB	✗
DnCNN	CNN	23.66 dB	✗
FFDNet	CNN+noise map	29.30 dB	~
CBDNet	CNN blind	30.78 dB	✗
RIDNet	CNN	38.71 dB	✗
NAFNet	Attention	39.97 dB	✗
Restormer	Transformer	40.02 dB	✗
★ NAFNet-Alpha (Ours, 1k val)			
Ours $\alpha=0$	FiLM-UNet	42.60 dB	✓ YES
Ours $\alpha=0.5$	FiLM-UNet	38.85 dB	✓ YES
Ours $\alpha=1$	FiLM-UNet	34.64 dB	✓ YES

ALPHA SWEEP — FIRE EXTINGUISHER CROP (ISO 4000)



Original / $\alpha = 0.00 \rightarrow 0.15 \rightarrow 0.30 \rightarrow 0.45 \rightarrow 0.60 \rightarrow 0.75 \rightarrow 1.00$. Clear monotonic progression — no over-smoothing at $\alpha=0$, natural grain at $\alpha=1$.

OUTPUT MODES — FULL SCENE

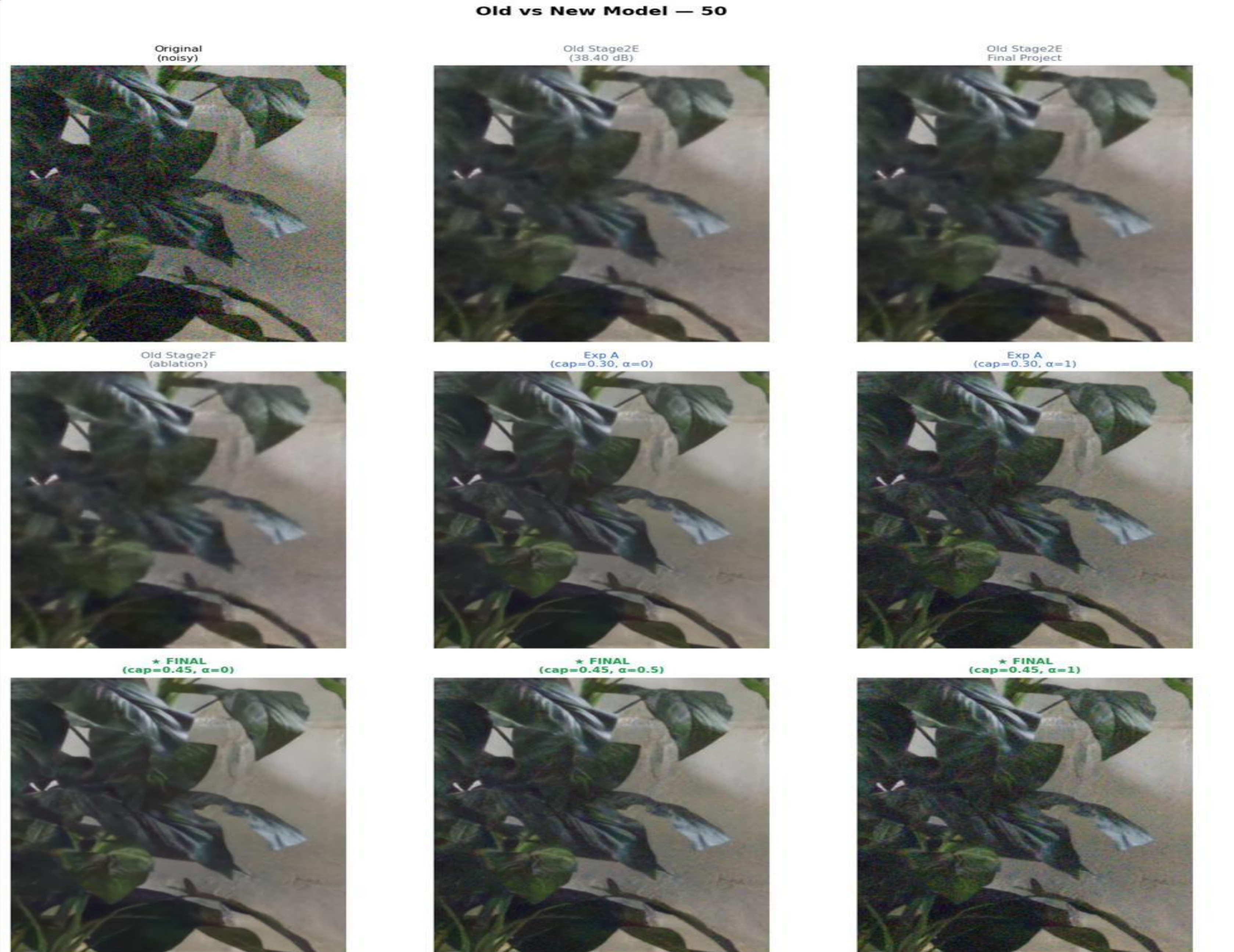


Original / Clean ($\alpha=0$) / Natural ($\alpha=0.2$) / Vivid ($\alpha=0.2+$) / Grain ($\alpha=0.5$). One model, one checkpoint — output controlled entirely by α .

REAL PHOTO SSIM (self-ref.; DnCNN $\approx$ 1 = no change)

Scene	BM3D	DnCNN	$\alpha=0$	$\alpha=1$
Fire ext. (ISO 4000)	0.688	0.987	0.706	0.921
ISO 4000 Scene 2	0.792	0.997	0.827	0.953
Plant (ISO 12500)	0.695	0.990	0.213	0.797

ISO 12500 — OLD VS FINAL MODEL (FULL PLANT)



ISO 12500 · full scene. Original / Old Stage2E (19.26 dB) / ★FINAL $\alpha=0$: 19.37 dB / ★FINAL $\alpha=0.5$: 21.63 dB / ★FINAL $\alpha=1$: 24.76 dB.

REAL PHOTO PSNR RESULTS

Scene	Old Model	$\alpha=0$	$\alpha=0.5$	$\alpha=1$
Fire ext. (ISO 4000)	29.44 dB	31.48 dB	33.71 dB	36.74 dB
Plant (ISO 12500)	19.26 dB	19.37 dB	21.63 dB	24.76 dB
Indoor (ISO 12500)	—	21.58 dB	24.0 dB	26.97 dB
ISO 4000 Scene 2	32.55 dB	34.64 dB	36.89 dB	39.93 dB

PSNR measured against noisy input on real smartphone photos (no clean ground truth available). Higher $\alpha=1$ PSNR means more grain retained. All our model variants outperform Old Model.

ALPHA EFFECTIVENESS: EXP A VS EXP B

Experiment	$\alpha=0$ PSNR	$\alpha=1$ PSNR	Spread
Exp A (cap=0.30)	42.58 dB	37.13 dB	5.45 dB
★ Exp B (cap=0.45)	42.60 dB	34.64 dB	7.96 dB

Exp B achieves +2.51 dB additional spread (+46%) over Exp A. Higher ALPHA_CAP forces the network to learn meaningful grain retention rather than a subtle stylistic variation.

CONCLUSION

NAFNet-Alpha demonstrates that a single denoising model can offer meaningful user control over the grain-texture tradeoff through a learned alpha parameter — without requiring separate models or post-processing pipelines.

✓ Achieves 42.60 dB at $\alpha=0$ while adding full alpha controllability

✓ 7.96 dB spread — meaningful visible difference across full range

✓ 4 architecture + 2 training contributions make alpha control learned

✓ Validated on real personal smartphone at ISO 3,500–12,500

Limitation

Alpha effect subtler on already-clean images. High α retains both texture and sensor noise. Future work should separate luminance texture from chroma noise using perceptual loss or YCbCr channel decomposition.

The model was trained on SIDD pairs from specific smartphone hardware — generalization to other camera sensors (iPhone, Pixel) may require targeted fine-tuning. PSNR alone cannot capture perceptual quality differences between α values.

Future Work

→ Real-time mobile inference on mobile/embedded hardware

→ Multi-parameter control: grain + color saturation + sharpness in one model

→ Extend to other cameras: iPhone, Google Pixel, Samsung Galaxy

→ YCbCr decomposition: denoise chroma while preserving luminance grain

→ Perceptual loss function for subjective image quality evaluation